Product Page (PDP)

How it works in Magento 2, and what our theme changes

Theme: Codilar/Training (parent Magento/blank) · Magento Open Source 2.4.9

Project: /var/www/html/training · Design reference: 2xlhome.com

Who this is for. Developers learning Magento 2 front-end on this training project. It explains the product detail page (PDP) end to end: how a URL becomes a page, which Magento folders take part, how the theme overrides them, and, file by file, everything the **Codilar/Training** theme changes to match the 2XL Home design.

How to use it. Read sections 1–4 for the concepts, keep section 5 (the annotated page) open while reading the code, and work through the exercises in section 11. Every path in this guide is relative to the project root `/var/www/html/training` unless it starts with /.

Item	Value
Example product	Tresa 8 Seater Dining Table · SKU 130917 · product ID 6
Example URL	http://localhost:8001/tresa-8-seater-dining-table.html
Theme folder	app/design/frontend/Codilar/Training/
Main PDP layout (ours)	Magento_Catalog/layout/catalog_product_view.xml
Main PDP styles (ours)	web/css/source/_catalog.less
Dev server command	php -S localhost:8001 -t pub phpserver/router.php

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1. What the PDP is made of

The product detail page is not one template. Magento assembles it at request time from three ingredients, and every change you make to the page is a change to one of them:

Ingredient	What it is	Where it lives	Changed by
Layout XML	The structure: which blocks and containers exist, their order and nesting.	`view/frontend/layout/*.xml` in modules and themes	*Merging*: theme files are applied on top of module files.
Blocks + templates	A block is a PHP object with data/logic; it renders one .phtml template.	`Block/` (PHP) and `view/frontend/templates/` (.phtml)	*Fallback*: a theme file with the same path replaces the module file.
Front-end assets	LESS → CSS, JavaScript (RequireJS modules), images, translations.	`web/`, `etc/view.xml`, `i18n/`	LESS is compiled per theme; JS/images use the same fallback as templates.

Rule. Golden rule: **never edit files under vendor/**. Composer overwrites them on every upgrade. Change behaviour in a theme (look and markup) or a module (PHP logic: plugins, observers, preferences).

2. Request lifecycle: from URL to HTML

Follow one request for `/tresa-8-seater-dining-table.html`:

```
Browser  GET /tresa-8-seater-dining-table.html
   │
   ▼
pub/index.php              (dev server: through phpserver/router.php)
   │   bootstraps Magento, creates the frontend application
   ▼
Front controller → routers, tried in order
   │   1. UrlRewrite router: looks the path up in table `url_rewrite`
   │        request_path  tresa-8-seater-dining-table.html
   │        target_path   catalog/product/view/id/6
   │   2. Standard router: catalog/product/view → Magento\Catalog\Controller\Product\View
   ▼
Controller  Product\View::execute()
   │   reads product id (6) and optional category id
   ▼
Helper  Magento\Catalog\Helper\Product\View::prepareAndRender()
   │   • loads the product (ProductRepository) and checks it is visible/enabled
   │   • stores it in the registry  → every block can call $block->getProduct()
   │   • initProductLayout(): adds layout handles (below)
   ▼
Layout engine
   │   • collects every XML file named after each handle, in ALL modules + theme chain
   │   • merges them into one tree, applies <move>, remove="true", arguments
   │   • wraps the tree in the page layout (1column, 2columns-left …)
   ▼
Rendering   each block → toHtml() → its .phtml template → HTML
   ▼
Browser     RequireJS boots the JS declared in data-mage-init / text/x-magento-init
            (validate-product, catalogAddToCart, our js/qty-stepper, Magento modals …)
```

Layout handles added for this product

A *handle* is just a name; Magento loads every layout file with that name. The PDP adds several, from general to specific, which is what lets you customise *all* products, one *type*, or a *single SKU*:

Handle	File name to create	Applies to
default	default.xml	every page of the store (header, footer)
catalog_product_view	catalog_product_view.xml	every product page
catalog_product_view_type_simple	catalog_product_view_type_simple.xml	simple products only (also _configurable, _bundle …)
catalog_product_view_attribute_set_4	…_attribute_set_4.xml	products using attribute set 4 (Default)
catalog_product_view_id_6	…_id_6.xml	only the product with entity ID 6
catalog_product_view_sku_130917	…_sku_130917.xml	only SKU 130917 (Tresa)

The code that adds them, from `vendor/magento/module-catalog/Helper/Product/View.php` (`initProductLayout()`):

```
$resultPage->addPageLayoutHandles(['type' => $product->getTypeId()], null, false);
$resultPage->addPageLayoutHandles(['attribute_set' => $product->getAttributeSetId()], null, false);
$resultPage->addPageLayoutHandles(['id' => $product->getId(), 'sku' => $urlSafeSku]);
```

> **Why the router matters.** Because the page is built from the URL rewrite table, the product URL only works when the request reaches Magento. On the PHP built-in server that requires `phpserver/router.php`; without it category and product URLs return 404 and CSS/JS under /static/version... return 404 (unstyled page).

3. The layout system in practice

Layout XML describes a tree of two kinds of nodes:

- **Container**: a wrapper with no PHP class. It only groups children and can output an HTML tag (`htmlTag="div" htmlClass="…"`). Example: `product.info.main`.
- **Block**: a PHP class + a template. Example: `product.info.addtocart` = `Magento\Catalog\Block\Product\View` + `product/view/addtocart.phtml`.

The instructions you will use most

Instruction	Effect	Example from our PDP
<referenceContainer name="X">	Open an existing container to add children to it.	add Tabby/Tamara, warehouse, accordion to product.info.main
<referenceBlock name="X">	Open an existing block (change arguments, add children).	add a "new_label" child to catalog.product.related
<block class=... template=... />	Create a new block.	product.info.bnpl → bnpl.phtml
<container name=... />	Create a new wrapper.	pdp.top, product.info.heading
<move element= destination= />	Re-parent an existing element.	move page.main.title into product.info.heading
remove="true"	Remove an element (and its children).	product.info.mailto, view.addto.compare
before= / after=	Order among siblings; "-" means first/last.	before="-" puts the gallery first in pdp.top
<arguments>	Pass data into a block (getData()).	page_size=5 for Recently Viewed

The core PDP tree (simplified)

From `vendor/magento/module-catalog/view/frontend/layout/catalog_product_view.xml`, plus blocks other modules add. Names in this tree are what you reference in your own XML:

```
content
├── product.info.main                (container, div.product-info-main — right column)
│   ├── page.main.title               (moved here by Luma; Blank keeps it above the columns)
│   ├── product.info.price            (container)
│   │   ├── product.info.stock.sku    (container: stock status + SKU)
│   │   ├── product.info.review       ("Be the first to review this product")
│   │   └── product.price.final       (price box → product/price/final_price.phtml)
│   ├── alert.urls
│   ├── product.info                  (the <form id="product_addtocart_form">)
│   │   └── product.info.form.content
│   │       └── product.info.addtocart  (qty + Add to Cart → product/view/addtocart.phtml)
│   │            └── (Braintree, Instant Purchase, PayPal buttons are added as children)
│   ├── product.info.extrahint
│   │   └── product.info.social → product.info.addto (wishlist, compare), product.info.mailto
│   └── product.info.overview         (short description)
├── product.info.media                (container, div.product.media — left column)
│   └── product.info.media.image      (gallery → product/view/gallery.phtml, Fotorama)
└── product.info.details              (tabs: product.info.description, product.attributes, reviews.tab)
content.aside
├── catalog.product.related          (Related products → product/list/items.phtml, type related)
└── product.info.upsell              (Up-sells → product/list/items.phtml, type upsell)
```

Lesson learned. A <move>d element cannot be used as the anchor for `before/after` of other new blocks in the same container: the move happens after the blocks are placed. We hit this: the badge and SKU first rendered at the bottom of the column. The fix was a dedicated container (`product.info.heading`) with the new blocks declared in order, then moving the title/SKU in relative to those blocks.

4. Theme fallback and the folders involved

Our theme declares <parent>Magento/blank</parent> in `theme.xml`. When Magento needs a file it searches this chain:

```
Template  Magento_Catalog::product/view/gallery.phtml
  1. app/design/frontend/Codilar/Training/Magento_Catalog/templates/product/view/gallery.phtml   ← our
     theme (wins)
  2. vendor/magento/theme-frontend-blank/Magento_Catalog/templates/product/view/gallery.phtml    ←
     parent theme
  3. vendor/magento/module-catalog/view/frontend/templates/product/view/gallery.phtml           ←
     module
  4. vendor/magento/module-catalog/view/base/templates/…                                        ←
     module, all areas
```

File type	Fallback behaviour
.phtml templates, JS, images, fonts	**Replaced**: the first file found in the chain is used, the rest are ignored.
Layout XML (*.xml in layout/)	**Merged**: all files with the handle name are combined; theme files are applied last. (Files in `layout/override/` replace instead; avoid them.)
LESS	`_extend.less` is **added** to the parent styles; `_theme.less` **replaces** the parent's variable file.
etc/view.xml	Image sizes and gallery options; our copy is used instead of Blank's.
i18n/en_US.csv	Translations; theme strings override module strings.

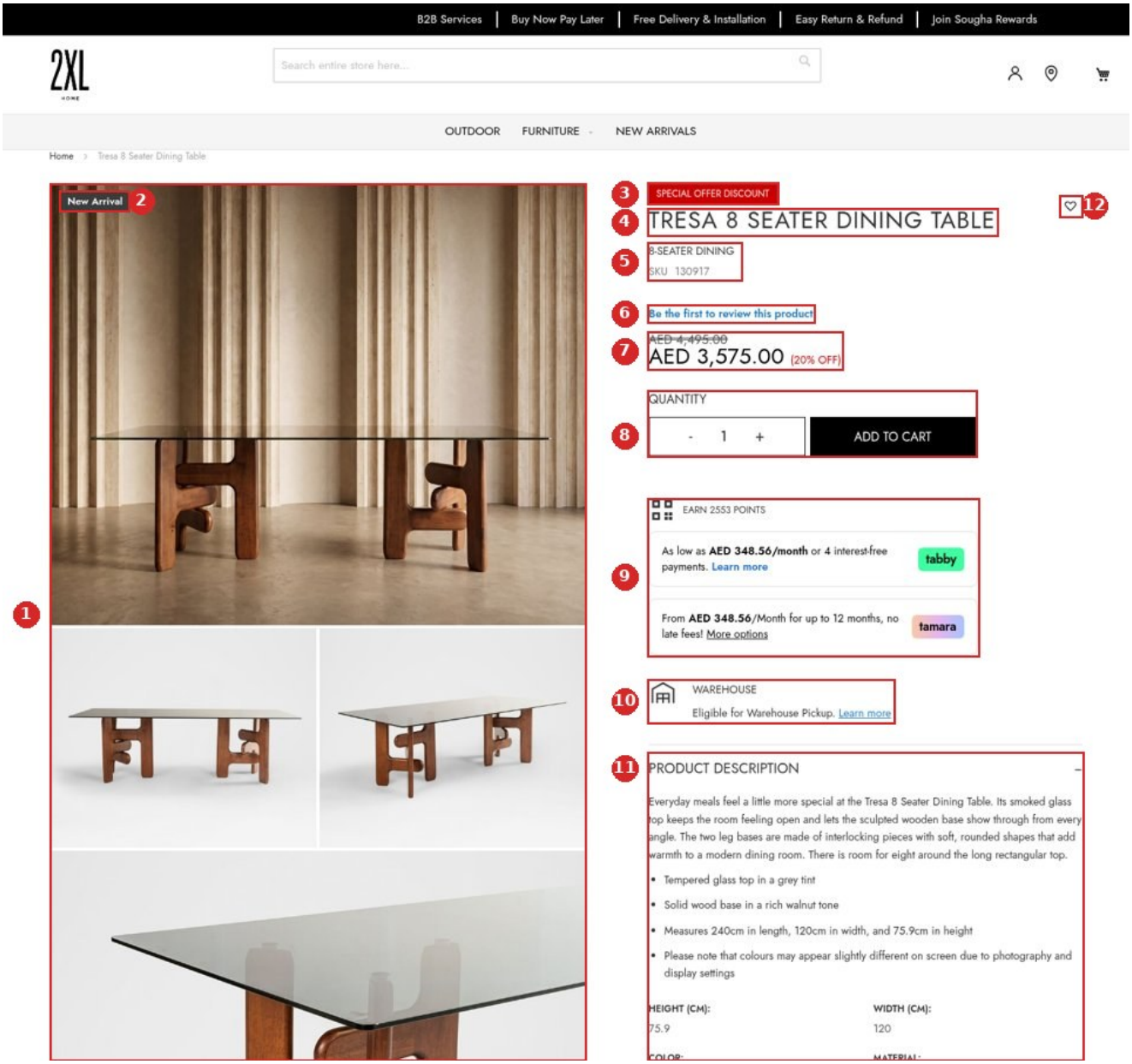
Core folders that build the PDP

Purpose	Path (under vendor/magento/)
Controller (entry point)	module-catalog/Controller/Product/View.php
Loads product, adds handles	module-catalog/Helper/Product/View.php

Purpose	Path (under vendor/magento/)
Blocks	module-catalog/Block/Product/View.php, View/Gallery.php, View/Description.php, View/Attributes.php, ProductList/Related.php, ProductList/Upsell.php
Layout	module-catalog/view/frontend/layout/catalog_product_view.xml (+ _type_simple, _type_virtual)
Templates	module-catalog/view/frontend/templates/product/view/*.phtml, templates/product/list/items.phtml
Price templates	module-catalog/view/base/templates/product/price/*.phtml
JS	module-catalog/view/frontend/web/js/ (catalog-add-to-cart.js, validate-product.js); lib/web/mage/gallery/ (Fotorama)
Parent theme styles	theme-frontend-blank/Magento_Catalog/web/css/source/
Other modules on the PDP	module-review (reviews), module-wishlist (heart), module-configurable-product + module-swatches (options), module-bundle, module-downloadable, module-product-video, module-instant-purchase, module-paypal, module-braintree

5. The finished page, part by part

Our PDP for Tresa (desktop, 1440 px). Each number maps to a block and the file that renders it.



#	Part	Block name (layout)	Template / source	Origin
1	Photo grid	product.info.media.image	product/view/gallery.phtml	core block, **our template**
2	New Arrival label	product.info.new.label	product/new-label.phtml	**new**
3	Special Offer badge	product.info.badge	product/view/badge.phtml	**new**
4	Title	page.main.title	Magento_Theme html/title.phtml	core, **moved**
5	Category + SKU	product.info.category, product.info.stock.sku	product/view/category.phtml; core sku	**new** + core moved
6	Review summary	product.info.review	core review summary	core
7	Price + (x% off)	product.price.final	product/price/final_price.phtml	core block, **our template**
8	Quantity −/+ and Add to Cart	product.info.addtocart	product/view/addtocart.phtml + web/js/qty-stepper.js	core block, **our template**
9	Points, Tabby, Tamara	product.info.bnpl	product/view/bnpl.phtml	**new**
10	Warehouse	product.info.warehouse	product/view/warehouse.phtml	**new**
11	Description, specs, shipping	product.info.accordion	product/view/accordion.phtml	**new** (core Attributes block)
12	Wishlist heart	product.info.addto	core addto.phtml (styled)	core, CSS only
–	Service icons	product.info.services	product/view/services.phtml	**new**
–	Complete the Collection	product.info.upsell	product/list/items.phtml	core block, **our template**
–	Related Products	catalog.product.related	product/list/items.phtml	core block, **our template**
–	Recently Viewed	product.recently.viewed	core widget grid.phtml	**added** by our layout

All template paths above are inside `app/design/frontend/Codilar/Training/Magento_Catalog/templates/` when marked "our template" or "new". The bottom of the page:





HEIGHT (CM): 75.9

WIDTH (CM): 120

COLOR: Grey

MATERIAL: Tempered glass & Solid wood

LENGTH (CM): 240

CARE: Wipe with clean & soft cloth

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6. Our layout file, explained

`Magento_Catalog/layout/catalog_product_view.xml` is merged on top of the core file. Below it is shown in full; the table after it explains each group.

```
<?xml version="1.0"?>
<page xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
      xsi:noNamespaceSchemaLocation="urn:magento:framework:View/Layout/etc/page_configuration.xsd">
    <body>
        <!-- Photos + details side by side in their own wrapper, so the sticky
             details column stops at the end of the photos -->
        <referenceContainer name="content">
            <container name="pdp.top" htmlTag="div" htmlClass="pdp-top" before="-"/>
        </referenceContainer>
        <move element="product.info.media" destination="pdp.top" before="-"/>
        <move element="product.info.main" destination="pdp.top" after="-"/>

        <!-- Badge, title, category, SKU grouped at the top of the right column -->
        <referenceContainer name="product.info.main">
            <container name="product.info.heading" htmlTag="div" htmlClass="pdp-heading" before="-">
                <block class="Magento\Catalog\Block\Product\View" name="product.info.badge"
                       template="Magento_Catalog::product/view/badge.phtml"/>
                <block class="Magento\Catalog\Block\Product\View" name="product.info.category"
                       template="Magento_Catalog::product/view/category.phtml"/>
            </container>
        </referenceContainer>
        <move element="page.main.title" destination="product.info.heading" after="product.info.badge"/>
        <move element="product.info.stock.sku" destination="product.info.heading" after="product.info.
    category"/>

        <referenceContainer name="product.info.main">
            <!-- Sougha points + Tabby / Tamara instalments (between Add to Cart and Warehouse) -->
            <block class="Magento\Catalog\Block\Product\View" name="product.info.bnpl"
                   template="Magento_Catalog::product/view/bnpl.phtml" after="product.info"/>
            <block class="Magento\Framework\View\Element\Template" name="product.info.warehouse"
                   template="Magento_Catalog::product/view/warehouse.phtml" after="product.info.bnpl"/>
            <block class="Magento\Catalog\Block\Product\View\Attributes" name="product.info.accordion"
                   template="Magento_Catalog::product/view/accordion.phtml" after="product.info.warehouse"/>
        </referenceContainer>

        <referenceContainer name="product.info.media">
            <block class="Magento\Catalog\Block\Product\View" name="product.info.new.label"
                   template="Magento_Catalog::product/new-label.phtml" before="-"/>
        </referenceContainer>

        <!-- Service icons under the accordion in the right column -->
        <referenceContainer name="product.info.main">
            <block class="Magento\Framework\View\Element\Template" name="product.info.services"
                   template="Magento_Catalog::product/view/services.phtml" after="product.info.accordion"/>
        </referenceContainer>

        <!-- 2XL shows no short description -->
        <referenceBlock name="product.info.overview" remove="true"/>

        <!-- Description + specs live in the accordion; the tabs keep only Reviews -->
        <referenceBlock name="product.info.description" remove="true"/>
        <referenceBlock name="product.attributes" remove="true"/>

        <!-- Below the product: Complete the Collection (up-sells), Related Products, Recently Viewed -->
        <move element="product.info.upsell" destination="content.aside" before="-"/>
        <referenceContainer name="content.aside">
            <block class="Magento\Catalog\Block\Widget\RecentlyViewed" name="product.recently.viewed"
                   template="Magento_Catalog::product/widget/viewed/grid.phtml" after="-">
                <arguments>
                    <argument name="uiComponent" xsi:type="string">widget_recently_viewed</argument>
                    <argument name="page_size" xsi:type="number">5</argument>
```

```
                    <argument name="show_attributes" xsi:type="string">name,image,price</argument>
                    <argument name="show_buttons" xsi:type="string"></argument>
                </arguments>
            </block>
        </referenceContainer>

        <!-- "New Arrival" badge on the Related / Complete the Collection cards -->
        <referenceBlock name="catalog.product.related">
            <block class="Magento\Framework\View\Element\Template" name="catalog.product.related.new.label"
                   as="new_label" template="Magento_Catalog::product/new-label.phtml"/>
        </referenceBlock>
        <referenceBlock name="product.info.upsell">
            <block class="Magento\Framework\View\Element\Template" name="product.info.upsell.new.label"
                   as="new_label" template="Magento_Catalog::product/new-label.phtml"/>
        </referenceBlock>

        <!-- 2XL has no reviews section on the product page -->
        <referenceBlock name="product.info.details" remove="true"/>

        <!-- Keep only the wishlist heart -->
        <referenceBlock name="product.info.mailto" remove="true"/>
        <referenceBlock name="view.addto.compare" remove="true"/>
    </body>
</page>
```

Group	Why
pdp.top container + two moves	Puts gallery and details in one flex wrapper. The details column is `position: sticky`; sticky stops at the end of its parent, so the parent must contain only these two columns. Without it, the column slid over the content below.
product.info.heading	Groups badge, title, category and SKU in a fixed order (see the "moved anchor" lesson in section 3).
bnpl / warehouse / accordion / services	New blocks chained with after="…" so they appear in order under the add-to-cart form.
product.info.new.label	Label on the gallery; reuses the same template as the listing and home carousels.
remove="true" lines	Short description, the description/specs tabs and the Reviews section are not part of the 2XL design; mail-to-friend and compare links removed.
upsell move + Recently Viewed	Up-sells are shown first as "Complete the Collection"; the Recently Viewed widget is a core Magento widget block added with its arguments.
new_label children	items.phtml asks for a child block called `new_label`; declaring it here keeps the template free of hard-coded block creation.

7. Templates

Two kinds: **overrides** (a copy of a core template with targeted edits; same path, so fallback picks ours) and **new templates** (referenced only by our layout). When you override, copy the core file first and change as little as possible: on a Magento upgrade you must compare your copy with the new core version.

7.1 Overrides of core templates

Our file	Core original	What we changed
product/view/gallery.phtml	module-catalog/…/product/view/gallery.phtml	Fotorama slider removed; all images output as a grid of links (one wide, two side by side). Zoom-on-hover is gone; click opens the large image.
product/view/addtocart.phtml	module-catalog/…/product/view/addtocart.phtml	Label "Quantity"; − and + buttons around the input; JS widget attached with data-mage-init.
product/price/final_price.phtml	module-catalog/view/base/…/price/final_price.phtml	Adds (20% off) when there is a special price. Used on PDP, listing, carousels.
product/list/items.phtml	module-catalog/…/product/list/items.phtml	Up-sell: Add to Cart shown; Related: "check items to add" checkboxes off; New Arrival label rendered from the new_label child.

gallery.phtml (ours, complete)

```
<?php
$productName = $block->getProduct()->getName();
$width = $block->getImageAttribute('product_page_image_medium', 'width');
$height = $block->getImageAttribute('product_page_image_medium', 'height');
$position = 0;
?>
<div class="pdp-gallery">
    <?php foreach ($block->getGalleryImages() as $image): ?>
        <a class="pdp-gallery-item" href="<?= $escaper->escapeUrl($image->getData('large_image_url'))
    ?>"
           target="_blank" rel="noopener">
            <img src="<?= $escaper->escapeUrl($image->getData('large_image_url')) ?>"
                 alt="<?= $escaper->escapeHtmlAttr($image->getLabel() ?: $productName) ?>"
                 width="<?= (int)$width ?>" height="<?= (int)$height ?>"
                 <?php if ($position++ === 0): ?>itemprop="image"<?php else: ?>loading="lazy"<?php
    endif; ?>/>
        </a>
    <?php endforeach; ?>
</div>
```

addtocart.phtml: the edited part

```
<div class="field qty">
            <label class="label" for="qty"><span><?= $block->escapeHtml(__('Quantity')) ?></span></
    label>
            <div class="control qty-stepper" data-mage-init='{"js/qty-stepper": {}}'>
                <button type="button" class="qty-btn" data-qty-step="-1"
                        aria-label="<?= $block->escapeHtmlAttr(__('Decrease quantity')) ?>">-</button>
                <input type="number"
                       name="qty"
                       id="qty"
                       min="0"
                       value="<?= $block->getProductDefaultQty() * 1 ?>"
                       title="<?= $block->escapeHtmlAttr(__('Qty')) ?>"
                       class="input-text qty"
                       data-validate="<?= $block->escapeHtml(json_encode($block->getQuantityValidators(
    ))) ?>"
                       />
                <button type="button" class="qty-btn" data-qty-step="1"
                        aria-label="<?= $block->escapeHtmlAttr(__('Increase quantity')) ?>">+</button>
            </div>
        </div>
```

final_price.phtml: the added discount label

```
    <?php
    $regularValue = $priceModel->getAmount()->getValue();
    $discount = $regularValue > 0
        ? round(($regularValue - $finalPriceModel->getAmount()->getValue()) / $regularValue * 100)
        : 0;
    ?>
    <?php if ($discount > 0) :?>
        <span class="price-discount">(<?= (int)$discount ?>% <?= $block->escapeHtml(__('off')) ?>)</
    span>
    <?php endif; ?>
```

7.2 New templates

File	Block class used	Renders
product/view/badge.phtml	Catalog\Block\Product\View	"Special Offer Discount" when final price < regular price
product/view/category.phtml	Catalog\Block\Product\View	Category the shopper came from, else the deepest category
product/new-label.phtml	any block with getProduct()	"New Arrival" while today is inside Set Product as New From/To
product/view/bnpl.phtml	Catalog\Block\Product\View	Points line, Tabby + Tamara boxes, two Magento modals with plans
product/view/warehouse.phtml	Framework\View\Element\Template	Static warehouse pickup note
product/view/accordion.phtml	Catalog\Block\Product\View\Attributes	Description + specs (getAdditionalData()) and Shipping and Returns in <details>
product/view/services.phtml	Framework\View\Element\Template	Four circle icons

Tip. Pick the block class by what the template needs. `Product\View` gives `getProduct()`; `View\Attributes` adds `getAdditionalData()` (all attributes with "Visible on Catalog Pages on Storefront" = Yes); a plain `Template` is enough for static content.

accordion.phtml (ours, complete)

```
<?php
/** @var \Magento\Framework\Escaper $escaper */
$product = $block->getProduct();
$outputHelper = $this->helper(\Magento\Catalog\Helper\Output::class);
$specs = $block->getAdditionalData();
?>
<div class="pdp-accordion">
    <details open>
        <summary><?= $escaper->escapeHtml(__('Product Description')) ?></summary>
        <div class="pdp-accordion-content">
            <?= /* @noEscape */ $outputHelper->productAttribute($product, $product->getDescription(),
    'description') ?>
            <?php if ($specs): ?>
                <dl class="pdp-specs">
                    <?php foreach ($specs as $spec): ?>
                        <div class="pdp-spec">
                            <dt><?= $escaper->escapeHtml($spec['label']) ?>:</dt>
                            <dd><?= /* @noEscape */ $outputHelper->productAttribute($product, $spec[
    'value'], $spec['code']) ?></dd>
                        </div>
                    <?php endforeach; ?>
                </dl>
            <?php endif; ?>
        </div>
    </details>
    <details>
        <summary><?= $escaper->escapeHtml(__('Shipping and Returns')) ?></summary>
        <div class="pdp-accordion-content">
            <p><?= $escaper->escapeHtml(__('The delivery schedule will depend on the item and the time
    when you placed the order.')) ?></p>
            <p><?= $escaper->escapeHtml(__('The standard delivery for accessories is 2-3 working days
    and for furniture, it takes 5-7 working days.')) ?></p>
            <a href="#"><?= $escaper->escapeHtml(__('Learn more')) ?></a>
        </div>
    </details>
</div>
```

Good practice. Accordion sections use native `<details>/<summary>`: they open and close without JavaScript and are keyboard accessible. Always escape output: `$escaper->escapeHtml()` for text, `escapeHtmlAttr()` for attributes, `escapeUrl()` for links; `/* @noEscape */` only for HTML that Magento already produced safely (e.g. the output helper).

new-label.phtml (shared by PDP, listing and carousels)

```
<?php
/**
 * "New Arrival" label, shown while today is inside the product's Set Product as New From/To dates.
 * Used on the product page and on each product card in the listing.
 *
 * @var \Magento\Framework\View\Element\Template $block  Must have a product (getProduct())
 * @var \Magento\Framework\Escaper $escaper
 */
$product = $block->getProduct();
$today = date('Y-m-d');
$from = $product->getNewsFromDate();
$to = $product->getNewsToDate();
$isNew = $from && substr($from, 0, 10) <= $today && (!$to || substr($to, 0, 10) >= $today);
?>
<?php if ($isNew): ?>
    <span class="product-new-label"><?= $escaper->escapeHtml(__('New Arrival')) ?></span>
<?php endif; ?>
```

8. JavaScript on the PDP

Magento loads JS with RequireJS. A template does not include <script src>; it *declares* which module to run on which element, and Magento runs it after the page loads:

```
<!-- 1. data-mage-init on the element itself -->
<div class="control qty-stepper" data-mage-init='{"js/qty-stepper": {}}'> … </div>

<!-- 2. text/x-magento-init script, selector → module → config -->
<script type="text/x-magento-init">
    { "#product_addtocart_form": { "Magento_Catalog/js/validate-product": {} } }
</script>
```

A module path `js/qty-stepper` resolves to the theme file `web/js/qty-stepper.js` (served as `/static/…/js/qty-stepper.js`). The module returns a function that receives the config object and the element:

```
/**
 * -/+ buttons around the product page quantity input.
 */
define(['jquery'], function ($) {
    'use strict';

    return function (config, element) {
        var $input = $(element).find('input.qty');

        $(element).on('click', '[data-qty-step]', function () {
            var step = parseInt($(this).data('qty-step'), 10),
                value = Math.max(1, (parseInt($input.val(), 10) || 1) + step);

            $input.val(value).trigger('change');
        });
    };
});
```

JS on our PDP	Declared in	Purpose
js/qty-stepper (ours)	addtocart.phtml	− / + change the qty input, minimum 1
Magento_Ui/js/modal/modal (core)	bnpl.phtml	Tabby / Tamara popups; option trigger opens on click of [data-trigger=…]
Magento_Catalog/js/validate-product (core)	addtocart.phtml	validates qty/options, then catalogAddToCart submits by AJAX
widget_recently_viewed UI component (core)	catalog_product_view.xml	Recently Viewed list, rendered client-side from browser storage

Note. The Fotorama gallery JS (`mage/gallery/gallery`) is no longer initialised because our gallery.phtml does not declare it. That is how a template override can remove behaviour as well as markup.

9. Styles, image sizes and text

9.1 How the LESS is organised

File	Role
web/css/source/_theme.less	Variables only (replaces Blank's): base font Jost.
web/css/source/_extend.less	Entry point for our CSS; imports the partials below; header, full-width layout, menu.
web/css/source/_catalog.less	Prices, product listing cards, **everything on the PDP** (selector `.catalog-product-view`), Tabby/Tamara modals.
web/css/source/_footer.less, _home.less	Footer and home page.

Magento compiles two stylesheets: `styles-m.css` (all screens) and `styles-l.css` (desktop, loaded with media="min-width: 768px"). You choose with two guards from Magento's LESS library:

```
& when (@media-common = true) {          // → styles-m.css, every screen size
    .catalog-product-view .pdp-badge { … }
}

.media-width(@extremum, @break) when (@extremum = 'min') and (@break = @screen__m) {
    .catalog-product-view .product-info-main { position: sticky; top: 20px; width: 42%; }   // → styles-
    l.css
}
```

9.2 Specificity lessons from this PDP

Symptom	Cause	Fix used
Old price as big as the new price	Magento puts class `price-final_price` on the *old* price container too, so a rule for `.price-final_price .price` hit both.	Target `.special-price .price` and `.price-box > .price-container .price` only.
Qty box and button misaligned	Blank's desktop CSS (styles-l, loaded after styles-m) adds padding-top: 25px to `.box-tocart .actions` with the same specificity.	More specific selector (`.product-info-main .box-tocart`).
Button 10 px too high	Empty Braintree / Instant Purchase containers after the button add height.	`.actions { display: contents };` extra children get their own full-width row.
Product cards squeezed	Blank sets `width: 25%/33%` per card in styles-l; our grid already sizes them.	`width: auto !important` (same as Blank's margins).
Details column over the content below	Sticky element's parent also contained later sections.	New `pdp.top` container (layout), flex in CSS.

9.3 Image sizes: etc/view.xml

Every product image on the storefront is requested by an *image ID*; `view.xml` sets its size. 2XL photos are 1000×820 (landscape), Blank's defaults are square/portrait, so we changed these IDs:

Image ID	Used for	Size now
product_page_image_medium / product_page_main_image	PDP gallery	700 × 574
product_page_image_small	thumbnails	120 × 98

Image ID	Used for	Size now
category_page_grid	listing cards	400 × 328
related_products_list, upsell_products_list	Related, Complete the Collection	400 × 328
recently_viewed_products_grid_content_widget	Recently Viewed	400 × 328
new_products_content_widget_grid	home carousels	400 × 328

> **Remember.** After changing view.xml, delete the resized-image cache `pub/media/catalog/product/cache` (or run `bin/magento catalog:images:resize`), otherwise old sizes keep being served.

9.4 Text: i18n/en_US.csv

Core strings can be renamed without touching templates. Our theme changes the up-sell title:

```
"We found other products you might like!","Complete the Collection"
```

10. What lives in the admin, not in the theme

Part of what you see on the PDP is data and configuration. Knowing which is which saves a lot of searching:

On the page	Where it is set	Notes
Specs (Height, Width, Length, Color, Material, Care)	Stores → Attributes → Product (codes dimension_height, dimension_width, dimension_length, finish_color, material, care)	Shown because "Visible on Catalog Pages on Storefront" = Yes; read by accordion.phtml via getAdditionalData().
Complete the Collection / Related	Catalog → Products → product → Related Products, Up-Sells and Cross-Sells	Up-sell = Complete the Collection; Related = Related Products.
New Arrival label	Product → "Set Product as New From / To"	Dates; no code change needed to add/remove a label.
Special Offer badge and (x% off)	Product → Advanced Pricing → Special Price	Computed from regular vs final price.
Currency AED	Stores → Configuration → General → Currency Setup	Base, default and allowed currency = AED.
"Only registered users can write reviews"	Stores → Configuration → Catalog → Catalog → Product Reviews → Allow Guests to Write Reviews = No	
Category under the title	Product → Categories	Deepest category, or the one the shopper came from.
Product URL	Product → Search Engine Optimization → URL Key	Stored in url_rewrite; reindex if URLs do not work.

11. Working on the PDP: workflow, debugging, exercises

11.1 After each kind of change

You changed…	Run
A .phtml template	bin/magento cache:flush (developer mode: often visible on refresh)
Layout XML	bin/magento cache:flush
LESS	rm -rf pub/static/frontend/Codilar var/view_preprocessed && bin/magento cache:flush
JS in web/js	same as LESS (remove the published copy in pub/static)
etc/view.xml	rm -rf pub/media/catalog/product/cache && bin/magento cache:flush
i18n CSV	bin/magento cache:flush
A change never appears	rm -rf var/cache/* var/page_cache/* (always safe; Magento rebuilds them)

11.2 See which template renders what

```
php bin/magento dev:template-hints:enable && php bin/magento cache:flush
#   reload the PDP: every block shows its template path and class in red
php bin/magento dev:template-hints:disable && php bin/magento cache:flush
```

11.3 Common problems on this project

Symptom	Cause	Fix
Page shows as plain unstyled text	Dev server started without the router	php -S localhost:8001 -t pub phpserver/router.php
Category/product URL gives 404 but home works	Same: no router, or url_rewrite missing	Router; then bin/magento indexer:reindex
"Address already in use"	A server already runs on that port	Use it, or stop it (ps … | grep "php -S"; kill PID)
Layout change not visible after cache:flush	Stale layout cache entries	rm -rf var/cache/* var/page_cache/*
LESS change not visible / page unstyled after edit	Old CSS in pub/static, or a LESS syntax error	Clear pub/static + view_preprocessed; check var/log/system.log

11.4 Exercises

- **One product only.** Create `Magento_Catalog/layout/catalog_product_view_sku_130917.xml` that removes `product.info.warehouse`. Flush cache; only Tresa loses the warehouse note.
- **Move a block.** Move `product.info.services` above the accordion with `<move element="product.info.services" destination="product.info.main" before="product.info.accordion"/>`.
- **New template.** Add a "Free delivery in Dubai" line under the price: new block in `product.info.main` with `after="product.info.price"` and a small .phtml.
- **Compare an override.** Diff our addtocart.phtml with the core one (`diff vendor/magento/module-catalog/view/frontend/templates/product/view/addtocart.phtml app/design/frontend/Codilar/Training/Magento_Catalog/templates/product/view/addtoca`

`rt.phtml`) and explain every changed line.

- **Data, not code.** Clear "Set Product as New From" on Tresa in the admin and confirm the label disappears from the PDP, listing and home carousel.

12. Appendix: every theme file that touches the PDP

File (in app/design/frontend/Codilar/Training/)	Type	Section
theme.xml, registration.php	theme declaration	4
Magento_Catalog/layout/catalog_product_view.xml	layout (merged)	6
Magento_Catalog/templates/product/view/gallery.phtml	override	7.1
Magento_Catalog/templates/product/view/addtocart.phtml	override	7.1
Magento_Catalog/templates/product/price/final_price.phtml	override	7.1
Magento_Catalog/templates/product/list/items.phtml	override	7.1
Magento_Catalog/templates/product/view/badge.phtml	new	7.2
Magento_Catalog/templates/product/view/category.phtml	new	7.2
Magento_Catalog/templates/product/view/bnpl.phtml	new	7.2
Magento_Catalog/templates/product/view/warehouse.phtml	new	7.2
Magento_Catalog/templates/product/view/accordion.phtml	new	7.2
Magento_Catalog/templates/product/view/services.phtml	new	7.2
Magento_Catalog/templates/product/new-label.phtml	new (shared)	7.2
web/js/qty-stepper.js	JS module	8
web/css/source/_catalog.less	styles	9
web/css/source/_theme.less, _extend.less	styles (global)	9
etc/view.xml	image sizes	9.3
i18n/en_US.csv	translation	9.4
Magento_Theme/layout/default.xml, default_head_blocks.xml	header/footer/font (every page)	–

Files in the theme that do **not** affect the PDP: `Magento_Catalog/layout/catalog_category_view.xml`, `Magento_Catalog/templates/product/list.phtml`, `…/product/list/addto/view.phtml`, `Magento_LayeredNavigation/templates/layer/view.phtml` (category page); `Magento_Cms/layout/cms_index_index.xml`, `Magento_CatalogWidget/templates/…/carousel.phtml`, `web/js/carousel.js`, `_home.less` (home page); `Magento_Newsletter/templates/subscribe.phtml`, `_footer.less` (footer).